☑ 1. Explain IEEE 802.11 System Architecture

IEEE 802.11 is a standard for **Wireless LAN (WLAN)** communication.

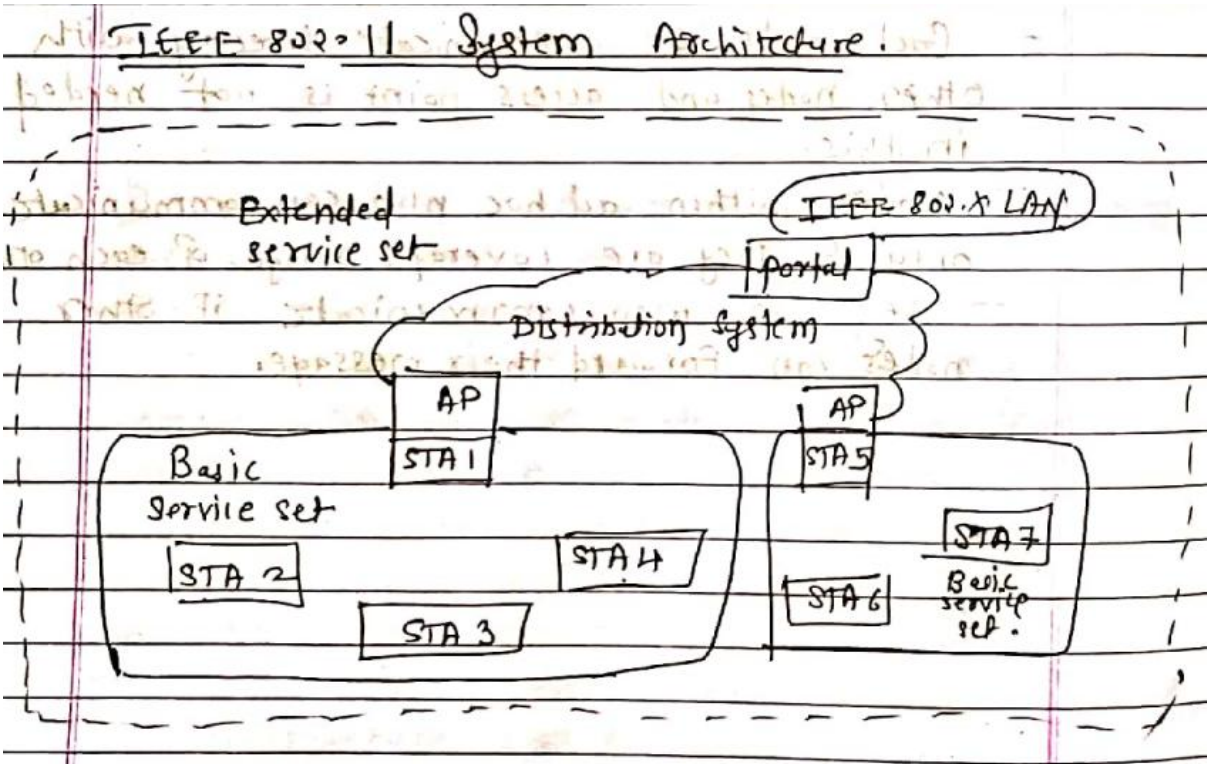
🔹 **Components:**

1. **Station (STA)**
   - Basic component of wireless network
   - Can be laptop, mobile, PC, etc.
   - Supports authentication, privacy, and data delivery
2. **Access Point (AP)**
   - Controls medium access
   - Connects wireless network to wired network
   - Acts as a bridge
3. **Basic Service Set (BSS)**
   - Group of stations connected to one AP
   - Communication is via AP (not direct)
4. **Extended Service Set (ESS)**
   - Multiple BSS connected using distribution system
5. **Distribution System (DS)**
   - Connects multiple BSS
   - Extends coverage area

🔹 **Working:**

- Stations communicate via **Access Point**
- AP connects to wired network
- Multiple BSS form ESS for large coverage

📊 **Diagram:**



☑ 2. Explain Topologies of Bluetooth

Bluetooth network forms two main topologies:

🔹 **1. Piconet**

- Basic Bluetooth network
- One device acts as **Master**
- Up to **7 active slaves**
- Communication controlled by master

🔹 **2. Scatternet**

- Multiple piconets connected together
- A device can act as **master in one** and **slave in another**
- Extends network coverage

🔹 **Features:**

- Short-range communication
- Low power consumption
- Used in personal area networks

☑ 3. Compare LTE and LTE Advanced

📍 From your notes: *Chapter 6 LTE introduction & evolution*

📌 **Answer:**

Feature	LTE	LTE-Advanced
Standard	4G (initial)	True 4G
Data Speed	Up to ~100 Mbps	Up to ~1 Gbps
Bandwidth	Limited	Carrier aggregation (higher bandwidth)
MIMO	Basic MIMO	Advanced MIMO (8x8, 4x4)
Performance	High	Much higher
Interference	Less optimized	Better interference handling

☑ 4. Explain SAE Architecture

SAE (System Architecture Evolution) defines **core network of LTE**.

🔹 **Main Components:**

1. **User Equipment (UE)**
   - Mobile device
   - Communicates with network
2. **E-UTRAN (Radio Access Network)**
   - Contains **eNodeB (eNB)**
   - Handles radio communication
3. **Evolved Packet Core (EPC)**
   Core network components:
   - **MME (Mobility Management Entity)** → control signaling
   - **S-GW (Serving Gateway)** → data routing
   - **P-GW (Packet Gateway)** → internet connectivity
   - **HSS (Home Subscriber Server)** → user database

🔹 **Features:**

- All-IP network

- Low latency
- High data rate
- Supports mobility

🔹 **Working:**

- UE connects to eNodeB
- eNodeB connects to EPC
- EPC connects to internet

📊 **Diagram:**

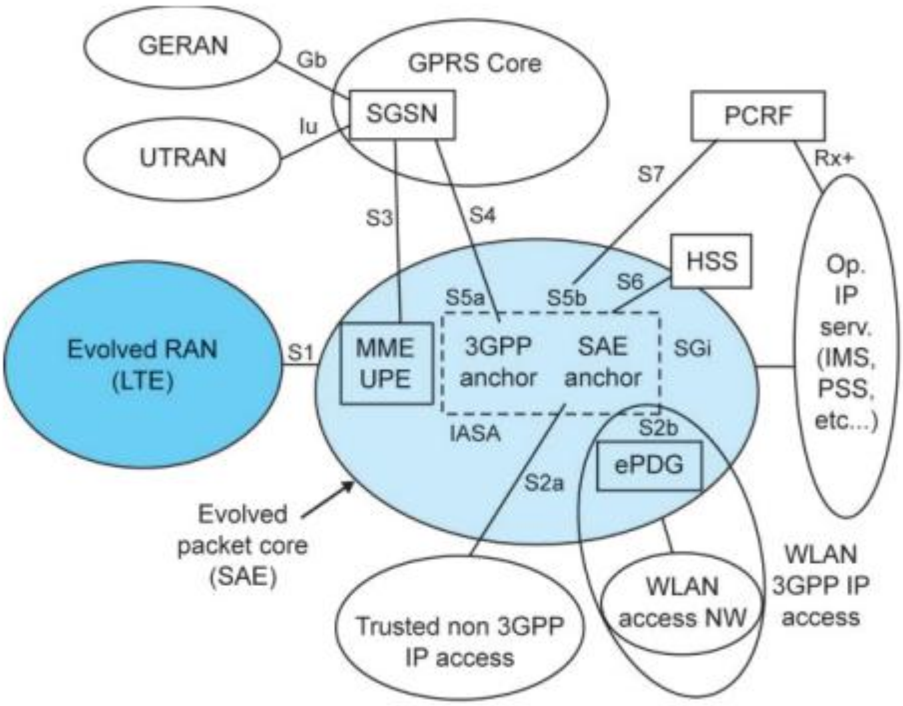


Figure 1.2-1. Logical high-level architecture for the evolved system (from 23.882 [2] Figure 4.2-1)

👉 *On page 2: High-level LTE/SAE architecture (UE, E-UTRAN, EPC*